Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

Louis HAUSEUX*, Raphaël ANTOINE°, Philippe FOUCHER°, Pierre CHARBONNIER°, and Josiane ZERUBIA*
*Inria, Université Côte d'Azur, Ayana team, France
°Cerema, ENDSUM research team, France
Corresponding author: louis.hauseux@inria.fr

***Abstract*—Automatic crack extraction is essential for both infrastructure inspection and geosciences, but annotated crack data remain scarce and domain shifts are frequent. We address the problem of extracting a crack network from one or several scalar modalities, without training.**

To do so, we generalize the classical FRANGI vesselness filter from a pixel-wise operator to a pairwise spatial graph. While the standard filter is limited to independent pixel-wise geometry, our formulation enriches this description by explicitly encoding the orientation alignment between neighboring pixels, alongside elongation and contrast. The final crack network is then extracted through a graph pipeline combining minimum spanning tree reduction and similarity-weighted tree centrality.

While supervised models perform best on clean, in-distribution data, our geometry-driven framework degrades more slowly than CrackSegDiff under the tested synthetic perturbations.

***Index Terms*—Crack detection, multi-modal fusion, FRANGI filter, graph, minimum spanning tree, tree centrality**

I. INTRODUCTION

Reliable crack mapping is needed for civil infrastructure inspection and for monitoring geological hazards such as landslides, faults and subsidence [1]. In practice, many crack maps are still produced or corrected manually, which is slow, subjective and difficult to scale. Automatic extraction is difficult because cracks are thin, discontinuous, and are often low-contrast structures whose appearance varies with the sensor, material, illumination, surface roughness and acquisition geometry.

Deep learning methods have become the dominant approach for crack segmentation. They perform well when training and test images follow the same distribution, but they require annotated data and may fail under sensor or domain shifts [2]–[4]. This limitation is especially relevant for multi-modal crack analysis, where public benchmarks with clean, well-aligned visible, depth or thermal modalities are still rare.

This paper takes a complementary direction: our framework provides an interpretable, training-free extraction pipeline. It addresses multi-modal scenarios for which annotations are not available. In this setting, creating ground truth is very expensive. Our method can directly assist domain experts (*e.g.,* geophysicists) and accelerate dataset annotation.

Our main contributions are as follows: (1) We generalize the FRANGI filter [5] from a pixel-wise vesselness response to a pairwise graph similarity using second-order shape, contrast and alignment terms, together with normalized multi-modal Hessian fusion; (2) We evaluate our complete algorithm on the FIND benchmark [6] and on two geological image sets, including comparisons with supervised models on clean and noisy data. The contribution lies in coupling these established components through the pairwise FRANGI graph.

II. RELATED WORK

A. Classical Curvilinear Filters

Multi-scale Hessian filters are a standard tool for enhancing curvilinear structures. The classical FRANGI vesselness filter [5] estimates whether a pixel belongs to a tubular structure from the eigenvalues of a smoothed Hessian matrix, following related work such as SATO's line filter [7]. These methods are fast and interpretable, but they are local: they evaluate pixels independently and can fragment under noise or textured backgrounds. Our method keeps the Hessian geometry but moves the decision from isolated pixels to edges of a spatial graph. Our preliminary study [8] considered two single-modal geological images. Here we add normalized multi-modal fusion, evaluate clean and noisy FIND data, and include a second geological setting.

B. Deep Crack Segmentation

Deep learning architectures such as U-Net [9], DeepCrack [10], CrackFormer [11] and recent diffusion models such as CrackSegDiff [12] have achieved strong crack segmentation results. Their main practical weakness is their dependence on annotations. A recent survey by ZHANG *et al.* [4] also emphasizes poor cross-domain robustness when sensors, materials or acquisition conditions change. This motivates training-free alternatives and hybrid strategies that can operate when annotations are not available.

C. Multi-Modal Fusion

In learned systems, multi-modal fusion is usually implemented by channel concatenation, intermediate feature fusion or late prediction fusion. These choices depend on the network architecture and on the modalities available at training time. Here, fusion is performed at the operator level: Each modality contributes a normalized Hessian, and the fused Hessian is processed by the same graph pipeline regardless of the number of modalities.

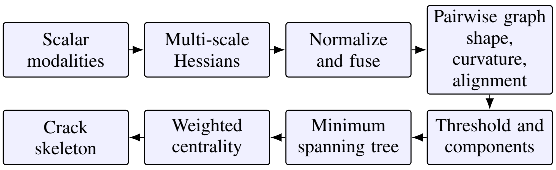


Fig. 1: Method overview. Local Hessian cues are fused across modalities and lifted to pairwise graph similarities; graph reduction and weighted centrality return the crack skeleton.

III. METHODOLOGY

A. Problem Formulation and Skeleton Output

Let $\Omega \subset \mathbb{Z}^2$ be the image grid and let

$$\mathcal{I} = \{I^{(m)} : \Omega \to \mathbb{R}\}_{m\in\mathcal{M}} \tag{1}$$

be one or several scalar modalities, such as visible intensity, depth/range or thermal infrared images. The output is a set of pixels $\widehat{\Gamma} \subset \Omega$ representing the crack skeleton. We use "skeleton" in the standard image-analysis sense: A one-pixel-wide centerline representation that preserves the main connectivity and branches of a thin object; thick masks used for evaluation are skeletonized with the implementation reported in [13], [14].

The method assumes that the crack of interest is visible as a dark curvilinear ridge. If a sensor encodes cracks as bright ridges, the modality can be inverted before processing. Figure 1 summarizes the processing chain.

B. Multi-Scale Hessian Features

For a modality $I^{(m)}$ and a Gaussian scale σ, the smoothed Hessian at $x \in \Omega$ is

$$\mathcal{H}_\sigma^{(m)}(x) = \begin{pmatrix} \partial_{xx}(G_\sigma * I^{(m)})(x) & \partial_{xy}(G_\sigma * I^{(m)})(x) \\ \partial_{xy}(G_\sigma * I^{(m)})(x) & \partial_{yy}(G_\sigma * I^{(m)})(x) \end{pmatrix}. \tag{2}$$

Equivalently, the derivatives are computed by convolution with derivatives of the Gaussian kernel [5], [15],

$$\partial_{ab}(G_\sigma * I^{(m)}) = (\partial_{ab}G_\sigma) * I^{(m)}, \qquad a, b \in \{x, y\}. \tag{3}$$

Since the Hessian matrix in (2) is real and symmetric, we compute its eigenvalues $\lambda_1^\sigma(x), \lambda_2^\sigma(x)$, ordered as $|\lambda_1^\sigma(x)| \le |\lambda_2^\sigma(x)|$. The eigenvector associated with $\lambda_1^\sigma(x)$ gives the local longitudinal direction of the ridge. For an elongated ridge, transverse curvature dominates longitudinal curvature, so $R_B = |\lambda_1^\sigma|/|\lambda_2^\sigma|$ is near zero; for a blob-like structure, the two magnitudes are comparable and R_B approaches one. Classical FRANGI vesselness combines this geometry with curvature magnitude and sign at each pixel. Our graph retains these local cues while requiring compatible evidence and longitudinal directions at neighboring pixels.

For multi-modal inputs, each Hessian is normalized by the largest spectral norm [16] in the image,

$$\widehat{\mathcal{H}}_\sigma^{(m)}(x) = \frac{\mathcal{H}_\sigma^{(m)}(x)}{\max_{y\in\Omega} \left\|\left|\mathcal{H}_\sigma^{(m)}(y)\right|\right\|_2}, \tag{4}$$

and the fused Hessian is

$$\mathcal{H}_\sigma^{\text{fused}}(x) = \sum_{m\in\mathcal{M}} w_m \widehat{\mathcal{H}}_\sigma^{(m)}(x), \qquad w_m \ge 0, \quad \sum_m w_m = 1. \tag{5}$$

All eigenvalues and orientations used below are computed from (5).

We assume that the crack geometry is shared across modalities, whereas sensor-specific artifacts are largely uncorrelated. Under this assumption, averaging the normalized Hessians reinforces consistent structures and reduces modality-specific noise.

C. *Generalized* FRANGI *Similarity Graph*

We build a sparse spatial graph $G = (V, E)$ on candidate pixels $V \subseteq \Omega$. We denote the distance between two candidate pixels by

$$\rho_{ij} = \|x_i - x_j\|_2 . \tag{6}$$

Pixels $x_i, x_j \in V$ are connected when

$$0 < \rho_{ij} \le R. \tag{7}$$

The radius R allows short interruptions caused by noise or local loss of contrast to be bridged, while ρ_{ij} discourages long accidental jumps.

For a scale $\sigma \in \Sigma$, an edge (i, j) is assigned a similarity $S_{ij}^\sigma \in [0, 1]$ as a product of three soft constraints.

Elongation. Both endpoints must have a small FRANGI ratio:

$$S_{\text{shape}}^\sigma(i, j) = \exp\left[-\frac{1}{2}\left(\frac{R_B(x_i) + R_B(x_j)}{s_\text{s}}\right)^2\right]. \tag{8}$$

Positive curvature. For dark ridges, the useful transverse curvature is $C_\sigma(x) = \lambda_2^\sigma(x)\mathbf{1}_{\lambda_2^\sigma(x)>0}$. The intensity term is

$$S_{\text{int}}^\sigma(i, j) = 1 - \exp\left[-\frac{1}{2}\left(\frac{\sqrt{C_\sigma(x_i)C_\sigma(x_j)}}{s_\text{i}}\right)^2\right]. \tag{9}$$

The geometric mean makes the edge weak when either endpoint has weak curvature.

Orientation alignment. If $\theta^\sigma(x)$ is the angle of the longitudinal eigenvector, direction compatibility is modeled by

$$S_{\text{align}}^\sigma(i, j) = \exp\left[-\frac{1}{2}\left(\frac{\sin(\theta^\sigma(x_i) - \theta^\sigma(x_j))}{s_\text{a}}\right)^2\right]. \tag{10}$$

The sine accounts for the fact that crack directions are unoriented: θ and $\theta + \pi$ describe the same local axis.

We first take the strongest similarity across scales,

$$\overline{S}_{ij} = \max_{\sigma\in\Sigma} S_{\text{shape}}^\sigma(i, j) S_{\text{int}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j), \tag{11}$$

then include spatial distance in the graph similarity and dissimilarity:

$$S_{ij} = \max\{0, 1 - (1 - \overline{S}_{ij})\rho_{ij}\}, \qquad d_{ij} = 1 - S_{ij}. \tag{12}$$

Thus the graph edge is strong only when both pixels are elongated, have the correct curvature sign and share a compatible orientation. The factor ρ_{ij} in (12) regularizes the later tree extraction.

For FIND and the Palais des Papes, we use $s_{\text{s}} = 2$, $s_{\text{i}} = 1/4$, $s_{\text{a}} = 1/8$, $\Sigma = \{1, 3, 5, 7\}$ and $R = 3$. For the Vaches Noires images, the single-modal implementation uses $s_{\text{s}} = 1/2$, $s_{\text{i}} = 1/4$, $s_{\text{a}} = 1/8$, $\Sigma = \{1, 3, 5, 7, 9\}$ to cover wider visible fractures, with the same radius $R = 3$ as in the preliminary study.

D. Graph Extraction

The extraction pipeline has four steps. First, thresholding removes low-confidence elements. A single retention factor τ is applied to both graph elements: the strongest fraction τ of edges is retained according to S_{ij}; nodes are then ranked by their maximum incident similarity, and the strongest fraction τ is retained.

Second, connected components are obtained from the remaining graph. On FIND, where the annotation targets one main crack, we retain the largest component. When several independent networks are of interest, the next two steps are applied to each retained component.

Third, a minimum spanning tree $\mathcal{T}$ [17] is computed using the dissimilarity d_{ij}. The tree preserves the "cheapest" coherent connections and turns the locally thick response into a structure suitable for centerline extraction.

Fourth, we compute a similarity-weighted tree centrality, related to betweenness centrality [18]. Unlike general-graph betweenness [19], this score is computed in $O(|V|)$ with a breadth-first traversal. Each vertex receives the weight

$$w_i = \max_{j:(i,j)\in E} S_{ij}. \tag{13}$$

After rooting the tree, let $\mathcal{T}_i$ be the subtree below x_i, $m_i = \sum_{x_k \in \mathcal{T}_i} w_k$, and $M = \sum_k w_k$. We use

$$C_{\mathcal{T}}(x_i) = m_i(M - m_i). \tag{14}$$

The highest-scoring vertex becomes the tree root, and the subtree masses are recomputed from it. The non-root scores follow (14) and are normalized, while the selected root is assigned score one. A branch is stopped when a child falls below the relative or absolute cutoff. Pixels on the main axis therefore remain, while short peripheral branches are removed.

Figure 2 shows the intermediate maps for a FIND example. The positive-curvature map is close to a classical pixel-wise FRANGI response; the graph similarity is cleaner because it also enforces pairwise orientation compatibility.

E. Role of the Components

The summary in Table I makes explicit which failure mode each term is intended to control.

The sensitivity parameters were selected empirically, starting from our preliminary study [8]. We retained s_{i} and s_{a} and increased s_{s} from $1/2$ to 2 for FIND and the Palais des Papes, allowing more variation in the FRANGI ratio, which is less stable than the curvature and orientation terms.

The scale set, radius and retention factor are fixed per dataset. We set Σ from the expected crack-width range and R from the largest gap that should be bridged; τ controls the sparsity–recall trade-off. We use $\tau = 0.25$ for FIND and the Palais des Papes, and $\tau = 0.30$ for Vaches Noires, following the preliminary study [8].

TABLE I: Role of the components in the extraction chain.

Component	**Role in the extraction chain**
S_{shape}	Penalizes blob-like responses and favors elongated local structures.
S_{int}	Strong positive curvature; closest part to classical FRANGI evidence.
S_{align}	Penalizes locally incompatible directions and accidental bridges.
ρ_{ij}	Discourages long links between distant but visually similar pixels.
Weighted centrality	Selects pixels lying on the main centerline of the retained graph.

IV. EXPERIMENTS

A. Datasets and Protocol

The quantitative evaluation uses FIND [6], a public benchmark with aligned intensity (I) and range (R) images for road-crack detection (and a third modality, "filtered" F, which is a clean post-processing of the range modality). We compare against CrackSegDiff [12], a supervised diffusion model for multi-modal crack segmentation. For both compared methods, the same evaluation protocol is used: Dense masks are skeletonized before comparison and overlap metrics are computed with a fixed spatial tolerance.

We also report experiments on two geological image sets provided by Cerema. The first contains two annotated 512×512 visible UAV patches from the Vaches Noires landslide at Villers-sur-Mer (France). We call these evaluation patches `Test1` and `Test2`; they are separate from the 3760×2058 image used to adapt the U-Net baseline. The second set is a photogrammetric intensity-range pair acquired on the retaining wall of the Palais des Papes in Avignon (France).

B. Metrics

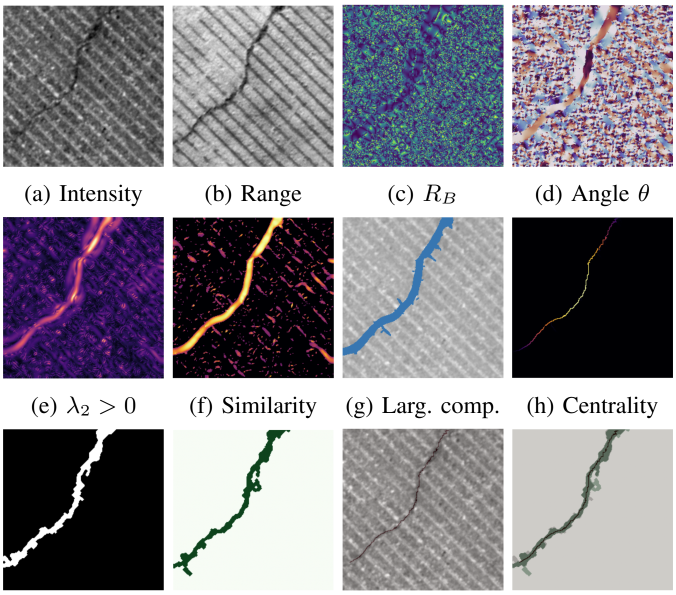
For a ground truth set A and a predicted set B, we use the JACCARD index [20], also known as Intersection over Union [21],

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \tag{15}$$

and the TVERSKY index [22],

$$T(A, B) = \frac{|A \cap B|}{|A \cap B| + \alpha\,|A \setminus B| + \beta\,|B \setminus A|}, \tag{16}$$

with $\alpha = 1$ and $\beta = 0.5$, so missed cracks are penalized more than over-detections. Since skeletons are one-pixel-wide, we dilate both skeletons by 3 pixels before computing these overlap scores. This tolerance is empirical and only compensates for small localization shifts at the 256×256 FIND resolution. We also report a discrete WASSERSTEIN distance [23], [24], which measures spatial displacement between predicted and reference skeleton pixels.



(a) Intensity (b) Range (c) R_B (d) Angle θ

(e) $\lambda_2 > 0$ (f) Similarity (g) Larg. comp. (h) Centrality

(i) Ground truth (j) CrackSegDiff (k) Our result (l) Overlay

Fig. 2: FIND example. **Top:** Intensity, Range, FRANGI ratio and orientation. **Middle:** Positive curvature, similarity, largest component and weighted centrality. **Bottom:** Ground truth, CrackSegDiff, ours (63% JACCARD, 71% TVERSKY, 8 px WASSERSTEIN) and overlay.

C. FIND: Clean and Noisy Data

Table II compares the proposed method with CrackSegDiff on clean FIND images using intensity and range. As expected, CrackSegDiff achieves the highest overlap on the clean, in-distribution data it was trained on. However, our geometric formulation still recovers the main topology of the fracture networks without requiring any training data, demonstrating its viability as a training-free baseline.

TABLE II: Performance on clean FIND data

Method	JACCARD	TVERSKY	WASSERSTEIN
Ours ($I+R$)	60%	68%	18 px
CrackSegDiff	**87%**	**90%**	**5 px**

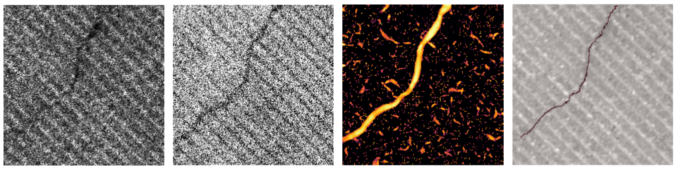
The FIND dataset [6] also contains a filtered range modality. Table III reports a modality contribution analysis for the proposed method. The filtered modality alone is strong, which shows the importance of clean, possibly pre-processed inputs for our method. Fusing intensity, range and filtered range gives the best result and supports the operator-level fusion strategy.

To evaluate the framework's robustness to synthetic sensor noise, we artificially degrade the input modalities. Specifically, we perturb the intensity modality with multiplicative speckle noise to simulate surface reflectance variations, and corrupt the range modality with additive Gaussian noise to model instrumental measurement errors:

$$I_{\text{noisy}}(x) = I(x) \cdot (1 + N(x)), \tag{17}$$

TABLE III: Modality contribution analysis for the proposed method on FIND. I: Intensity, R: Range, F: Filtered range.

Weights	JACCARD ↑	TVERSKY ↑	WASSERSTEIN ↓
$I = 1$	41%	48%	43 px
$R = 1$	54%	62%	20 px
$F = 1$	60%	68%	12 px
$I = \frac{1}{2}, R = \frac{1}{2}$	60%	68%	18 px
$I = \frac{1}{3}, R = \frac{1}{3}, F = \frac{1}{3}$	**63%**	**71%**	**11 px**



(a) Noisy Int. (b) Noisy Range (c) Similarity (d) Our result

Fig. 3: Noisy FIND example ($s = 0.5$) with speckle noise on intensity and Gaussian noise on range. The graph similarity remains concentrated around the main crack.

TABLE IV: Quantitative comparison on the two annotated Vaches Noires patches.

Image	Method	JACCARD ↑	TVERSKY ↑	WASSERSTEIN ↓
`Test1`	Ours	0.47	0.52	**34 px**
	U-Net + T.L.	**0.50**	**0.60**	38 px
`Test2`	Ours	0.29	0.32	81 px
	U-Net + T.L.	**0.34**	**0.36**	**60 px**

$$R_{\text{noisy}}(x) = R(x) + N'(x), \tag{18}$$

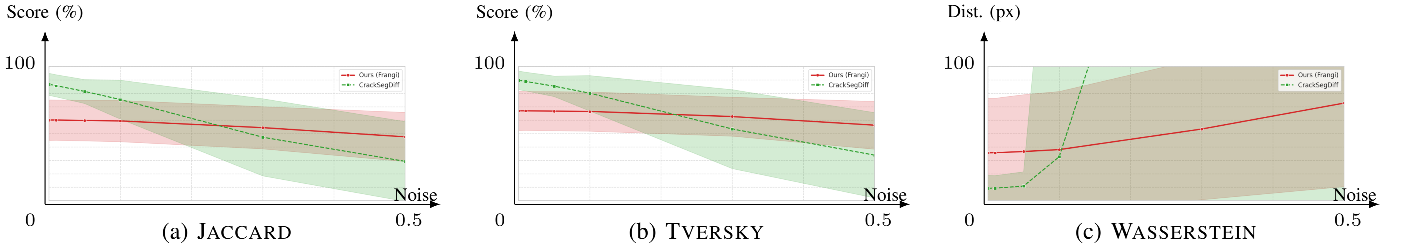
where $I(x)$ and $R(x)$ denote the original intensity and range signals at pixel x, respectively. $N(x), N'(x) \sim \mathcal{N}(0, s^2)$ are independent Gaussian random variables where s^2 measures the variance of the noise. This protocol evaluates the robustness of our operator-level fusion to controlled sensor perturbations.

Figure 3 shows an example at a strong noise level; Figure 4 reports the degradation curves. CrackSegDiff initially performs better, but its score decreases rapidly as noise increases. The proposed method starts lower but degrades more smoothly because the graph uses multi-scale curvature and orientation compatibility instead of learned appearance statistics alone.

D. Geological Domain Shift

The Vaches Noires data differ markedly from the road-crack FIND images and contain only a visible modality. We use $\Sigma = \{1, 3, 5, 7, 9\}$, $R = 3$ and $\tau = 0.3$ for both evaluation patches.

We compare against a U-Net baseline adapted by transfer learning [25] on an annotated 3760×2058 pixel image from the same area, following the common strategy used when only a small amount of domain-specific annotation is available. This ground truth annotation required approximately two days of work by an expert geophysicist. The proposed method uses only the visible image and no training annotation.



(a) JACCARD (b) TVERSKY (c) WASSERSTEIN

Fig. 4: Performance degradation on noisy FIND. Red: Proposed method; green: CrackSegDiff. Shaded regions show standard deviation. The horizontal axis is the noise level s (standard deviation of the Gaussians).

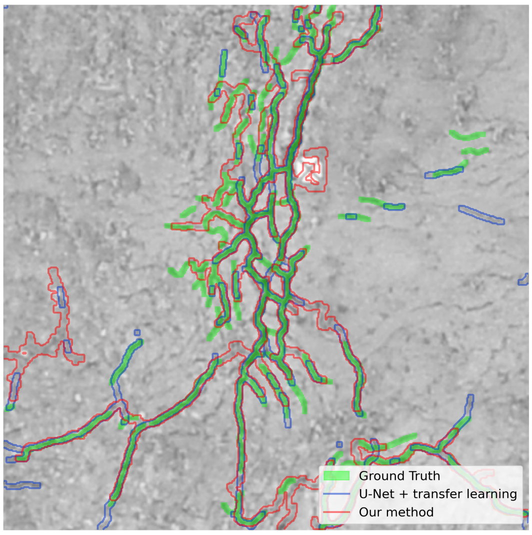


Fig. 5: Vaches Noires `Test1`. Ground truth: green; ours: red; U-Net + transfer learning: blue.

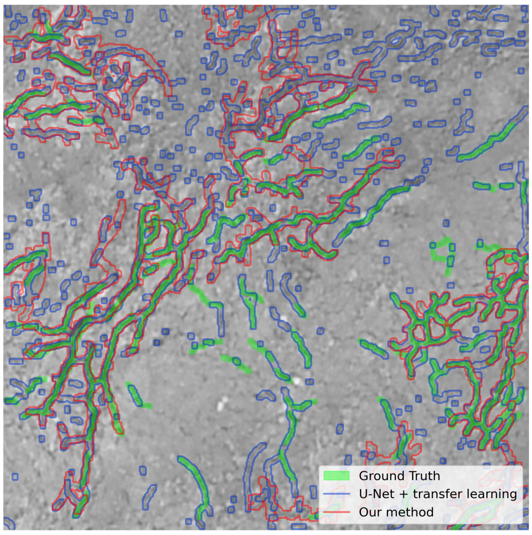
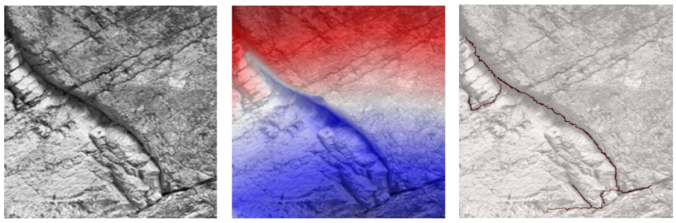


Fig. 6: Vaches Noires `Test2`. Ground truth: green; ours: red; U-Net + transfer learning: blue.

The supervised baseline gives slightly higher overlap scores on both patches, which is expected because it was adapted on a nearby annotated image. Our method uses no annotation and gives a lower WASSERSTEIN distance on `Test1`. These two patches are useful transfer tests, but they are not enough to establish performance across domains.

We also apply the method to a fractured limestone wall at the Palais des Papes, using an orthoimage and a photogrammetric depth map. Figure 7 shows that the main fracture is recovered without retraining; minor cracks are missed where depth resolution and contrast are insufficient.



(a) Intensity (b) Int. + Range (c) Ours

Fig. 7: Palais des Papes wall: intensity, depth overlay and extracted skeleton.

Code and Data Availability

The code and reproduction notebooks are available at https://github.com/Ayana-Inria/Frangi-EUVIP. The FIND benchmark is publicly available [6]. The non-public data used in our experiments are available at https://drive.google.com/drive/folders/1iC7QrdB2vZmaunjjPrIcu9r5wMeBPMlk.

V. Conclusion

We presented a geometric, training-free method for multi-modal crack skeletonization. On clean FIND data, the supervised model performs better; under added noise, our graph method degrades more slowly. The geological results show that the method can be applied without retraining, but the evaluation is limited to two annotated patches and one qualitative case.

The output is an editable spatial graph, which is useful for semi-automatic annotation. Vision foundation models are now essential baselines for crack segmentation; a direct comparison with SAM [26] and its crack-specific adaptation CrackSAM [27] is therefore a priority. We will also test sensitivity on established datasets [28] and higher-order graph variants [29].

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